L-TRACE: Lung Temporal Radiomic Analysis for Cancer Early Detection

Introduction

- Lung cancer is the leading cause of cancer-related mortality, with 5-year survival of 18%, largely due to late-stage diagnosis.
- Low-dose CT (LDCT) screening reduces mortality in high-risk smokers but has limitations, incl. a 96.4% false-positive rate.
- Traditional radiologic assessments rely on static imaging, while machine learning can analyze serial scans to detect changes in nodule progression, texture, and vascularization over time.
- In this study, we present L-TRACE, a deep learning approach integrating longitudinal LDCT scans with clinical features such as age, race, gender, and smoking history to improve lung cancer prediction before radiographic progression.
- This is the first study to apply deep learning feature extraction across time to LDCT, addressing a major gap in current screening. By using a unique cohort design, our model improves screening accuracy, reduces false positives, and enables earlier intervention.

Methods

- Scans from the National Lung Screening Trial (NLST) were used to develop and validate L-TRACE.
- The dataset included 747 cases (147 cancer-positive, 600 cancer-negative), with 42% (n=62) of positive cases classified as high-grade (G3/G4).
- Two cohorts were analyzed: (1) all 747 cases with two sequential LDCT scans for model training and validation, and (2) 147 cancer-positive cases for tumor grade prediction. This approach used longitudinal imaging to improve predictive accuracy and stage differentiation.

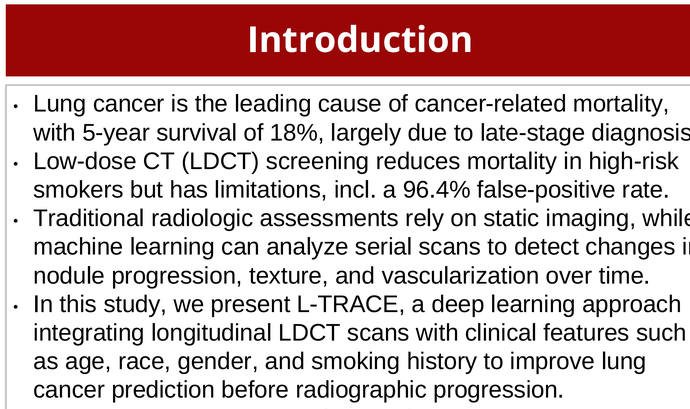


Figure 1: L-TRACE integrates a VGG16-based CNN for lung CT scans and an MLP for clinical data, combining features to predict lung cancer risk and prognosis.

- Training ran for 20 epochs with a 70:30 train-test split.
- Performance was assessed via AUC and F1-score. Saliency maps highlighted predictive regions. Prognostic capabilities, including tumor stage and grade were also evaluated.
- A Streamlit-based web app allows clinicians to upload LDCT scans, input clinical data, and receive real-time risk predictions.
- NLST data were obtained from The Cancer Imaging Archive, with planned validation using Henry Ford Health System data under an IRB-approved protocol. 100 cases (n=50 cancer, n=50 benign) will be used to assess generalizability.

Results

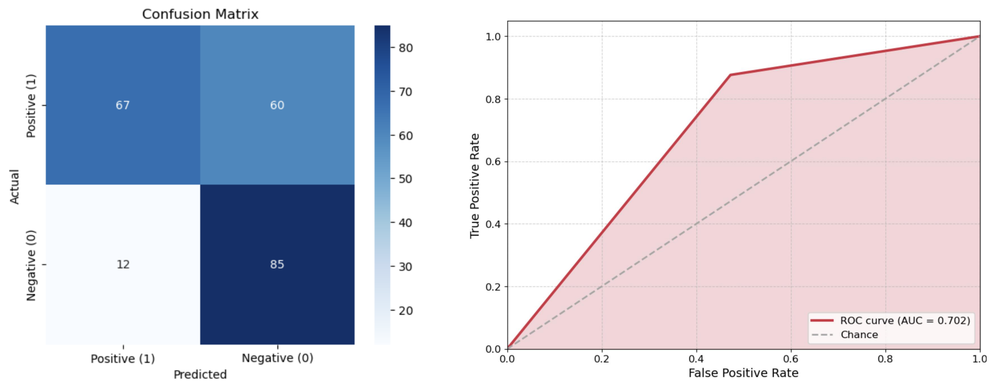


Figure 2: L-TRACE struggles with single-scan predictions, achieving moderate accuracy (0.679) and high recall (0.876) at the cost of low precision (0.586), leading to frequent false positives. AUC-ROC (0.702) highlights limitations of static imaging for reliable cancer detection.

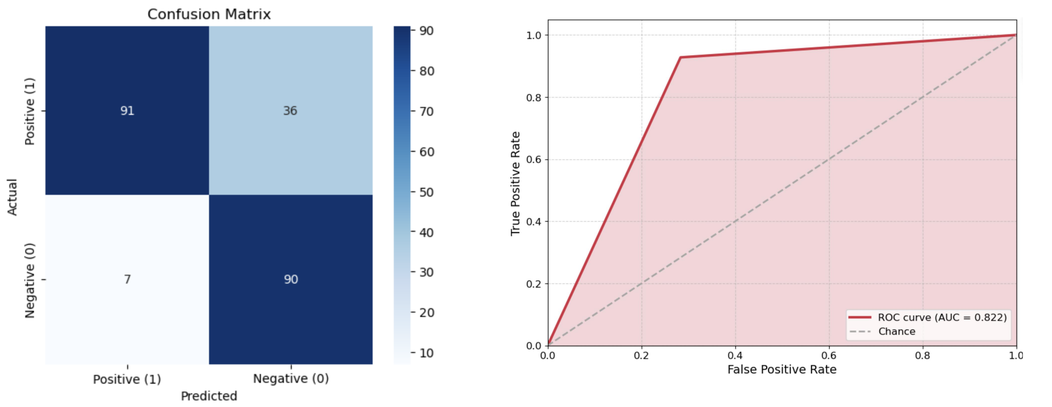


Figure 3: Integrating sequential LDCT scans and clinical data significantly improves performance, increasing accuracy (0.808) and AUC-ROC (0.822). Higher precision (0.714) and recall (0.928) indicate longitudinal imaging improves malignancy differentiation, reducing false positives while maintaining sensitivity.

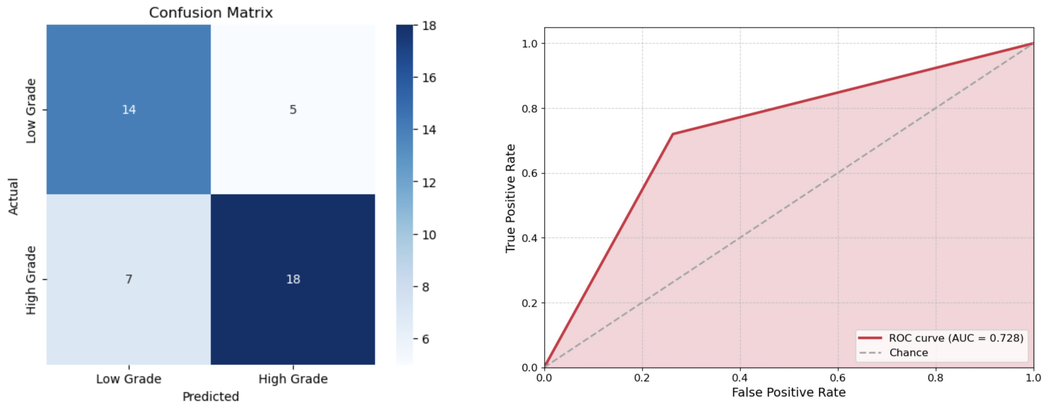


Figure 4: L-TRACE predicts histological grade with moderate accuracy (0.727) and an AUC-ROC of 0.728, demonstrating the potential for radiomics-based grading.

Photographs were created by student researcher.

Figure 5: Saliency map highlights nodule margins and vascular structures as imaging features driving lung cancer prediction. High-intensity regions indicate influence on model decision making.

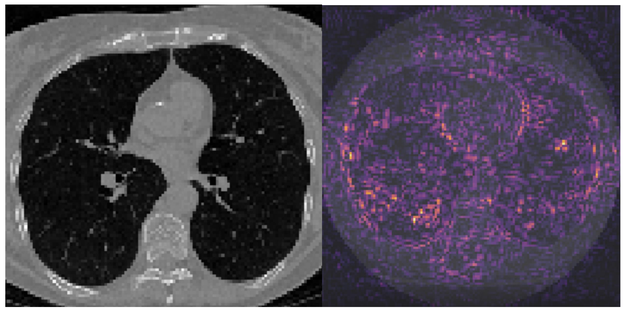
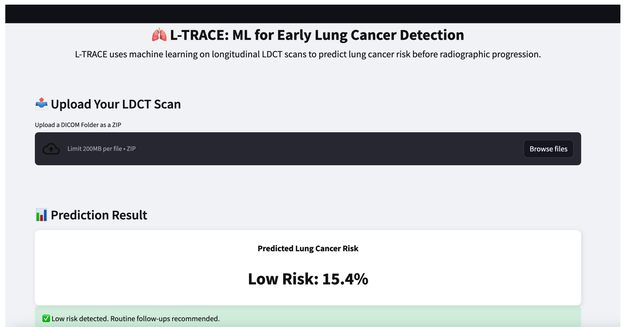


Figure 6: L-TRACE web app allows clinicians to upload LDCT scans for real-time risk assessment.



Conclusion

- This study is the first to demonstrate integrating sequential LDCT scans with clinical data improves lung cancer risk prediction (AUC: 0.822 vs. 0.702).
- By using longitudinal imaging, L-TRACE moves beyond static risk assessment, capturing tumor evolution over time and addressing a key limitation in screening models.
- It also demonstrates that ML can predict histological grade from LDCT scans (AUC: 0.728), supporting non-invasive, radiomics-driven tumor characterization.
- Saliency maps confirm that models prioritize biologically relevant imaging features.
- A Streamlit-based web application translates these findings, offering an accessible and interpretable tool.
- With planned external validation in 100 cases, this study highlights the benefits for integrating machine learning into routine lung cancer screening.

Works Cited

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